

Quantifying the Cost of Monitoring Record Fragmentation in Short Horizon PM_{2.5} Forecasting

Abstract—Short horizon PM_{2.5} forecasting is benchmarked almost entirely on accuracy, on records whose gaps are treated as a preprocessing detail. This paper asks whether the fragmentation of the monitoring record, rather than the model class, decides which method wins. Nine years of hourly observations from Dhaka (7.44 usable years, 82.3 per cent hourly coverage) are fused with reanalysis meteorology into 102 predictors, and compact sequence models under a 100,000 parameter budget are compared against tree ensembles, linear models and the classical baselines air quality work often omits, at horizons from 1 to 24 hours. At 24 hours the validation selected recurrent model (16,193 parameters) attains an RMSE of $57.70 \mu\text{g m}^{-3}$ against the best classical model at 58.50, with a Holm adjusted Diebold and Mariano p of 0.037; a 95 per cent model confidence set nonetheless retains five of nine candidates, so the finding is about tiers and not architectures. A controlled gap injection experiment then degrades a near complete record two ways that remove identical numbers of observed hours and differ only in arrangement. The sequence tier loses 0.0394 skill to arrangement alone (50 matched pairs, Holm $p < 0.0001$) and is the only family whose loss is significant on all three donor stations, although it is not separable from tree ensembles when families are differenced within matched triples. Because a scored row needs unbroken history, a gap costs the hours it removes plus the backward reach behind them. Shortening that reach from 192 to 48 hours removes 97 per cent of the penalty, and a closed form for usable rows fitted on 101 cells predicts 404 held out cells at $R^2 = 0.982$. Availability is therefore reported as a first class metric, since the configured reach answers only 76.4 per cent of test hours and shortening it costs no measurable accuracy.

Index Terms—air quality forecasting, PM_{2.5}, missing data, record fragmentation, recurrent neural networks, forecast availability, model confidence set

I. INTRODUCTION

Fine particulate matter is the environmental exposure most closely tied to premature mortality worldwide, and the World Health Organization guideline value is exceeded across most of South Asia for most of the year [1]. Dhaka is among the cities where a credible next day forecast would carry immediate public health value, and also among those where the observational record needed to build one is thinnest. Those two facts are usually treated as independent. They are not.

A decade of data driven air quality forecasting has converged on a familiar protocol. A series is obtained, absences are filled by interpolation or a nearest neighbour rule, a deep architecture is trained, and the error is compared against a small set of competitors [2], [3]. Missingness enters this protocol once, as a preprocessing decision, and leaves it as a sentence in the methods section. The literature treating missingness as a modelling object rather than a nuisance sits largely in clinical time series [4]. What has not been asked, to

our knowledge, is the prior question. If two records differ in how their absences are arranged, does the arrangement alone change which method a benchmark recommends?

The question cannot be settled by comparing cities, because two records drawn from different places differ in climate, instrument, span, pollution regime and continuity at once, so a ranking that moves between them is attributable to everything simultaneously. What is needed is a counterfactual on a single record. This paper builds one. We assemble a leakage controlled pipeline over nine years of hourly PM_{2.5} from Dhaka fused with reanalysis meteorology, benchmark three tiers of method at five horizons, and then hold the record fixed while manipulating only the arrangement of its absences. Fig. 1 shows the design. Our contributions are as follows.

- 1) **A controlled gap injection experiment** in which two arms remove identical hour counts and differ only in contiguity. Arrangement alone costs the sequence tier 0.0394 skill, the only family significant on all three donor stations.
- 2) **A mechanism, fixed by design rather than by regression.** A scored row needs unbroken history over a sterilisation radius; shortening it from 192 to 48 hours removes 97 per cent of the penalty.
- 3) **A closed form for usable supervision**, fitted on 101 cells and predicting 404 held out cells at $R^2 = 0.982$ across two further stations and a quartered radius.
- 4) **Forecast availability as a reported quantity**, with a decision rule for the backward reach that needs only run lengths and costs nothing to evaluate.

II. RELATED WORK

Recurrent architectures entered air quality forecasting by direct transfer from sequence modelling [5], [6], and the canonical demonstration on pollutant concentrations reports gains over feedforward and statistical competitors [2]. Subsequent work has broadened the architecture space without broadening the comparison much. Recent evidence that a single linear layer matches or exceeds transformer forecasters [7] makes the choice of baseline, rather than of architecture, the load bearing decision, which is why both proposed linear models enter here as first class competitors.

Missing data in air quality series is normally addressed as imputation, and comparative studies conclude by recommending one [3]. That framing assumes filled values restore the record, and under the classical taxonomy [8] the assumption concerns the mechanism generating absence. Our concern is downstream of it. Even when every filled value is correct, a

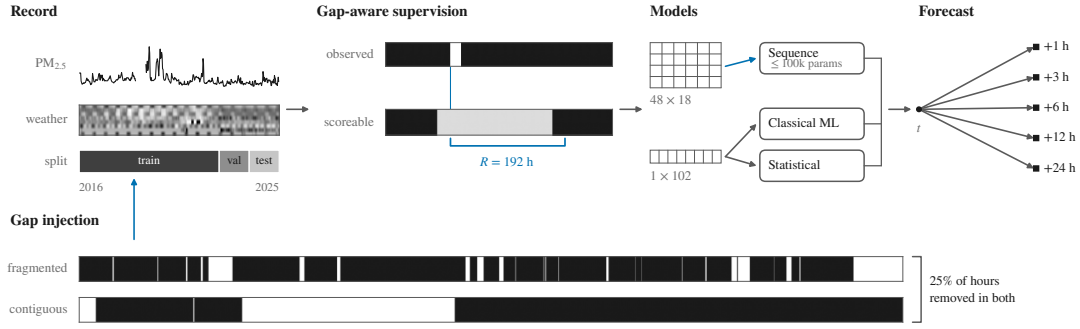


Fig. 1. Study design. A scored row requires unbroken history over a sterilisation radius R , drawn at the configured $R = 192$ h. Tabular models receive 102 engineered predictors per row; the sequence tier receives 18 contemporaneous channels over a window, because engineered history features restate what the window already contains. The lower panel shows the two injection arms, which remove the same number of observed hours and differ only in arrangement.

supervised row built from a window of history is unavailable whenever that window straddles a gap, so absences impose a structural cost no imputation policy removes. Work on recurrent models with informative missingness [4] shares the intuition but treats the pattern as a signal to exploit rather than a budget to spend.

The statistical apparatus for comparing forecasters is meanwhile mature and underused here. Pairwise accuracy tests [9], familywise error control [10], the model confidence set [11] and block resampling for dependent errors [12] are all standard, and their absence is why a table of closely spaced errors is read as a ranking.

III. STUDY DESIGN

Four components are described in turn below. Two hourly records supply the target and its drivers. A gap aware rule decides which rows may be scored at all, and it governs everything after it. Three tiers of method are then compared under a selection rule that never consults test error, and a controlled degradation makes fragmentation the manipulated variable.

A. Records and predictors

The primary target is hourly $\text{PM}_{2.5}$ from the United States Embassy monitor in Dhaka, retrieved from the OpenAQ archive [14], spanning 2016 to 2025. Of the 79,216 hours in that span, 65,214 are observed, giving 82.3 per cent coverage over 7.44 usable years, broken by 2,074 distinct gaps of which 1,324 last a single hour and the longest runs 2,712 hours. The distribution is severe, with a mean of $93.7 \mu\text{g m}^{-3}$ and half of all hours above the Bangladesh 24 hour standard of $65 \mu\text{g m}^{-3}$ [16]. Meteorological drivers come from NASA POWER hourly point reanalysis [15]. A second record, the Beijing multi site archive [13], supplies both the cross city comparison and the donor stations for the injection experiment.

Feature construction yields 102 predictors covering lagged concentrations, rolling statistics, meteorological history and calendar terms. The sequence tier does not receive them. It receives 18 contemporaneous channels over a window, because a lag feature at time t is by construction the same channel at

t minus the lag, already inside the window. Supplying both produced a heavily collinear input and handicapped the tier against competitors the same columns help.

B. What counts as a scored row

Five rules are enforced structurally rather than by convention. Meteorology is never read beyond the forecast origin. Splits are chronological and scalers are fitted on training data only. Imputation never crosses a split boundary and is limited to a short backward fill. Any window whose internal timestamps are not contiguous is rejected and counted. Each horizon is forecast directly by its own model rather than by recursive rollout.

The fourth rule organises the rest of this paper. A row at time t is scored only if it carries unbroken observed history back to $t - L$ and an observed target at $t + h$ in the same run. Writing W for the sequence window, the backward requirement is the *sterilisation radius*

$$R = \max(L, W - 1) + h, \quad (1)$$

which is 192 hours at the configured lookback and the 24 hour headline horizon. A gap therefore does not cost the hours it removes. It costs those hours plus the R behind them, which no longer reach back far enough to support a row.

C. Model tiers and the selection rule

Tier one comprises persistence, seasonal naive, climatology and SARIMAX. Tier two comprises ridge regression, random forest, XGBoost and LightGBM, tuned by randomised search over blocked time series cross validation [18]. Tier three comprises GRU, LSTM and the two linear forecasters of [7] under a 100,000 parameter budget [17], five seeds each, with two larger configurations excluded rather than the budget relaxed.

No model is selected on test error. Tier three is ranked by mean validation loss across seeds and tier two by cross validation score. Tier one is ranked on test error only because it has no hyperparameters and therefore nothing to bias. The rule has a measurable cost, which we report rather than assume away.

D. The gap injection experiment

A near complete donor record is degraded to five coverage levels by two arms. The *fragmented* arm removes many short outages whose lengths are drawn from the primary record’s own empirical gap length distribution. The *contiguous* arm removes the same number of hours as a few long blocks, spread evenly so the arms differ in contiguity and not in seasonal composition. The grid is 101 cells, five coverage levels by two arms by ten injection seeds plus an undegraded reference, and each cell is a full refit.

Three controls make the comparison mean what it claims. Both arms remove an identical hour count, so volume is held constant and no result is explained by there being less data. The evaluation period is never degraded, so every cell is scored on identical rows. The optimizer step budget, not the epoch budget, is equalised across cells, because fragmentation shrinks the training set and a fixed epoch count would leave a fragmented cell undertrained rather than damaged. Hyperparameters are frozen at the undegraded record’s selections, as is each family’s representative model, so a difference between arms cannot absorb a change of model.

E. Forecast availability

Every accuracy figure in this literature is conditional on the model being able to forecast at all, and that condition is rarely reported. We define forecast availability as the share of evaluation hours a configuration can answer. Because a model that declines the hard hours flatters itself on the hours it accepts, error on the served set is never compared across configurations of differing availability. The two valid comparisons are the common subset served by every arm and an all hours score, over a fixed universe, in which an unanswered hour falls back to persistence.

IV. RESULTS

Results follow the order the argument requires rather than the order of their strength. The benchmark comes first, because its limits motivate everything after it. The two city contrast follows and fails to identify anything, the controlled experiment then supplies what that contrast cannot, and the mechanism, the closed form and the decision rule close the section.

A. The benchmark, and what it can separate

Table I reports the 24 hour horizon. The validation selected recurrent model, a single layer GRU with 16,193 parameters over a 24 hour window, reaches an RMSE of $57.70 \mu\text{g m}^{-3}$ averaged over five seeds against 73.76 for persistence, a skill score of +0.218. The best classical competitor is random forest at 58.50, and the paired test on the selected seed rejects equality with a Holm adjusted p of 0.037. Intervals there are a moving block bootstrap with 1,000 resamples over 24 hour blocks, taken in blocks because consecutive hourly errors are correlated.

That is the strongest reading the data support, and it is weaker than it looks. The 95 per cent model confidence set retains five of the nine candidates, so the recurrent model,

TABLE I
TEST PERFORMANCE AT THE 24 HOUR HORIZON.

Model	RMSE	95% CI	Skill	MCS
GRU (16,193 par.)	57.24	[52.38, 62.17]	+0.218	yes
Random forest	58.50	[53.45, 63.49]	+0.207	yes
XGBoost	58.70	[53.64, 63.73]	+0.204	yes
LightGBM	59.02	[54.03, 64.19]	+0.200	yes
Ridge	59.37	[54.30, 64.34]	+0.195	yes
Climatology	65.43	[60.52, 70.45]	+0.113	no
Seasonal naive	73.76	[67.77, 79.86]	0.000	no
Persistence	73.76	[67.77, 79.86]	0.000	no
SARIMAX	81.36	[74.83, 87.93]	-0.103	no

RMSE in $\mu\text{g m}^{-3}$. The GRU row is the validation selected seed, whose predictions enter the paired tests; the five seed mean is 57.70 ± 0.44 .

the three tree ensembles and ridge are not separable at this horizon. The horizon is a choice rather than a finding. Across the five horizons tested, LightGBM beats the sequence tier significantly at 1 hour and at 12 hours, and the sequence tier wins significantly only at 24. Any single horizon headline is therefore a selection, and we report the four horizons that do not support ours next to the one that does.

The identity of the winner is also unstable. Re running the selection rule on 2,000 resamples of the seed set returns the reported configuration 45 per cent of the time and ten distinct candidates win at least once, yet the consequence is bounded, because the candidates validation cannot separate are near ties. Selecting on validation rather than on test costs $+0.05 \mu\text{g m}^{-3}$ here. We therefore make claims about tiers, not about architectures.

B. The observational contrast identifies nothing

Running the identical pipeline on the Beijing record gives a rank correlation between the two cities’ method orderings of only +0.20, so a benchmark run on either city alone would recommend a different method. The comparison is nevertheless uninformative about fragmentation, in two ways. First, the comparison city cannot support a ranking at all, because its own confidence set retains nine of nine candidates, persistence included. Second, the observational gradient runs the wrong way. The sequence tier ranks first on the more fragmented record and third on the near complete one, the opposite of what a fragmentation account predicts, and the pair also confounds fragmentation with a harder problem, a shorter evaluation period and a seasonal transition. We report this rather than set it aside, and treat it as the reason to build a counterfactual.

C. Arrangement costs skill at constant volume

Fig. 2 shows the experiment, and panel (b) is the mechanism in one line. At every coverage level the two arms have removed exactly the same number of observed hours, and yet the fragmented arm retains several thousand fewer usable supervised rows, a deficit that widens as the record degrades. Panel (a) shows what that costs. Each pair of a coverage level and an injection seed gives one matched observation, so each family contributes 50 pairs. We write the *arm gap* for fragmented minus contiguous skill within such a pair,

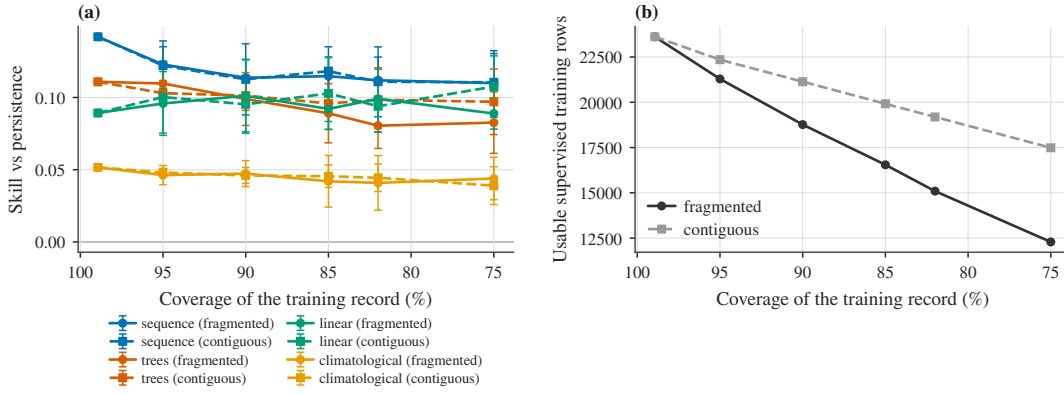


Fig. 2. The gap injection experiment on the donor record at 24 hours. (a) Skill against persistence by family, fragmented arm as circles and contiguous arm as squares, error bars over ten injection seeds. (b) Usable supervised training rows under the two arms. At every level the arms removed an identical number of observed hours, so the vertical distance in (b) is created purely by arrangement.

so a negative value means fragmentation costs that family more than removing the same hours contiguously. Table II reports the signed rank test, Holm corrected across families. The sequence tier loses 0.0394 skill to arrangement alone, and it is not alone in losing something. The climatological and tree families also move, and only the linear family is indistinguishable from zero, so the honest summary is that fragmentation is costly to most of the methods considered and most costly to the one that reads windows.

D. Replication, and the contrast that does not survive

One record cannot distinguish a property of fragmentation from a property of a station, so the experiment was repeated on two further stations of the same archive. We adopt a strict replication rule, requiring the same sign and Holm significance on every donor. Table II reports the outcome, an asterisk marking Holm significance within that donor. Only the sequence tier replicates. The climatological family is instructive here and is the reason more than one donor was run. On the first donor alone its penalty collapses to -0.096 at the severest degradation, which read in isolation would have argued that fragmentation degrades anything estimated from the record. It does not survive the other two stations, so that collapse is a property of that station.

Significance against a family's own null on every donor is not, however, a contrast between families. The stronger claim, that the sequence tier suffers *most*, requires differencing families within matched triples of coverage level, injection seed and donor. Under an intersection union test, correct here because the claim is a conjunction, it fails. The sequence tier separates from the linear and climatological families but not from the tree ensembles, where the difference is -0.0102 with $p = 0.15$. We therefore withhold the comparative claim and state the narrower one that survives, namely that the sequence tier's penalty is the only one appearing reliably across records and the largest in mean.

What does discriminate is the shape of the penalty in coverage. A cost flat in coverage is a fixed toll, whereas one that steepens is a mechanism. Fitting one slope per injection

seed and donor, the sequence penalty steepens by 0.0226 per ten percentage points of coverage lost while the tree penalty is flat and not significant. That is the shape the next subsection predicts, and the pooled test averages it away.

E. The backward reach carries the effect

The account in (1) predicts something falsifiable. If the penalty is caused by sterilised history rather than by lost hours, shortening the reach should shrink it. Because the reach is set by configuration rather than inferred from a regression, this is a stronger design than a regression based mediation and needs no ignorability assumption. The injection is deterministic in arm, coverage and seed and runs before any feature is built, so the same degraded series appears at every reach and each draw is a repeated measure.

The mediator moves first. Relative to the contiguous arm at an identical removed hour count, the fragmented arm loses 8,056 usable training rows at $R = 192$, 4,415 at $R = 72$ and 3,220 at $R = 48$. The penalty follows it, as Table III shows, the sequence gap falling from -0.0394 to -0.0012 . Each test there is one sided and declared in advance, because the mechanism predicts a single direction.

One confound must be bounded rather than argued away. A shorter reach scores more evaluation hours, 4,806 against 3,813, so the two gaps are not measured on the same rows. Even if those extra hours carried no arm difference whatever, the gap would shrink to 80 per cent of itself by arithmetic alone, which accounts for 0.0079 of the observed change of 0.0383 and leaves 79 per cent attributable to the reach. The falsification control holds too, since the linear family had no penalty to mediate and shows no mediation.

F. A closed form for usable supervision

If the radius is what a gap spends, the quantity of supervision a record can support should be predictable from the record alone. Writing U for usable rows, O for observed hours and k for distinct gaps,

$$U \approx O \exp(\beta_0 - \alpha Rk/O), \quad \alpha = 0.1555, \beta_0 = -0.1596. \quad (2)$$

TABLE II
EFFECT OF ARRANGEMENT AT CONSTANT DATA VOLUME.

Family	Pooled paired test			Per donor arm gap				Dose response	
	Mean gap	95% CI	p (Holm)	Wanliu	Dingling	Dongsi	Replicates	Slope	p (Holm)
Sequence	-0.0394	[-0.0565, -0.0236]	< 0.0001	-0.0394*	-0.0200*	-0.0273*	yes	-0.0226	0.016
Climatological	-0.0357	[-0.0609, -0.0138]	0.0115	-0.0357*	-0.0017	-0.0206*	no	-0.0247	0.062
Trees	-0.0263	[-0.0393, -0.0125]	0.0003	-0.0263*	-0.0120	-0.0179*	no	-0.0052	0.164
Linear	-0.0016	[-0.0081, +0.0047]	0.9542	-0.0016	-0.0356*	-0.0225*	no	-0.0173	0.019

Fifty matched pairs per family per donor; representatives are fixed on the undegraded record. Differencing families within matched triples does not separate sequence from trees (-0.0102, $p = 0.15$), so the comparative claim is withheld.

TABLE III
ARM GAP AT THE CONFIGURED REACH AGAINST A QUARTERED REACH.

Family	$R=192$	$R=48$	Change	Med.	p (Holm)
Sequence	-0.0394	-0.0012	+0.0383	97 %	< 0.0001
Climatological	-0.0357	-0.0005	+0.0351	99 %	0.0097
Trees	-0.0263	-0.0072	+0.0191	73 %	0.0017
Linear	-0.0016	+0.0021	+0.0037	—	0.2605

TABLE IV
AVAILABILITY AND THE PRICE OF THE BACKWARD REACH.

Record	Availability		Ampl. ($\times$)	Elast. at 192
	$R=192$	$R=48$		
Dhaka	76.4 %	87.2 %	2.3	0.372
Beijing Wanliu	72.5 %	85.5 %	34.2	0.139
Beijing Dingling	64.4 %	80.9 %	14.0	0.213
Beijing Dongsi	64.3 %	81.2 %	10.4	0.168

Fitted on the 101 cells of one donor grid at $R = 192$, (2) predicts 404 held out cells at $R^2 = 0.982$ with a median absolute error of 3.5 per cent, as Fig. 3 shows. Those cells are two different extrapolations. Two hundred and two come from the other stations at the same radius, asking whether the constants travel between records (R^2 of 0.970 and 0.983). The remaining 202 are the *fitted* station rebuilt at $R = 72$ and $R = 48$, asking whether R is a factor of the form or a scale the constants absorbed (R^2 of 0.979 and 0.964). Scoring that second group at the fitted radius instead collapses the pooled R^2 to 0.461, which is what the radius term carries.

The constant α is not free either. It should equal the share of gaps outliving the short forward fill, which is 0.1418 across this record's 2,074 gaps with a 95 per cent interval of [0.1274, 0.1574]. The fitted value falls inside that interval, so the constant is the imputation policy rather than a fitted parameter. The practical consequence is that a near complete record is not a well supervised one. Amplification, being usable hours destroyed per hour missing, runs at 2.3 on the primary record and at 34.2, 14.0 and 10.4 on the three near complete stations, because it follows the number of gaps and not their total length.

G. What the reach costs, and what shortening it buys

Anyone holding a record can price a reach exactly, with no fitted constants at all. A run of length l supports $\max(0, l - R)$ scored rows, so

$$U(R) = \sum_{\text{runs}} \max(0, l - R), \quad -\frac{dU}{dR} = \#\{l > R\}. \quad (3)$$

One further hour of backward reach costs exactly the number of runs still longer than it. Divided through by U/R this becomes an elasticity, the only form of the price comparable across records of different sizes. Table IV gives it. At the configured radius the primary record pays 0.372 against the donor's 0.139, so a reach one per cent deeper costs it 2.7 times as large a share of its supervision. The same table shows why coverage is the wrong summary statistic. Hourly coverage runs

82.3 per cent on the primary record against 98.9 on the donor, and yet availability runs 76.4 against 72.5, because the primary record's absences are clustered into a few long outages while the near complete stations' are scattered. Coverage is what a custodian reports. Availability is what a forecaster gets.

The trade was measured rather than assumed. Refitting every model at capped reaches and scoring on the 9,083 hours every arm can serve, no capped arm beats the configured reach and none loses to it, across 28 Holm corrected paired tests. Shortening the reach is not better at forecasting. It is better at answering. Over a fixed universe in which an unanswered hour falls back to persistence, capping at 24 hours raises availability from 83.5 to 97.6 per cent and is worth +0.0154 median skill, +0.0204 for the best model.

Two qualifications keep this honest. The gain is not the rescue of catastrophic hours, because the unserved hours are *easier* than the served ones, persistence scoring 56.90 there against 73.76. It is a model beating persistence over a large block of ordinary hours. And the free price sizes the opportunity without promising the payoff, since the primary record recovers 48 per cent of its training rows against the donor's 9 per cent and gains less skill. Availability rather than rows orders the two correctly.

V. DISCUSSION AND LIMITATIONS

The result that survives every check here is not a claim about architectures. It is that the arrangement of absences is an experimental variable with a measurable effect on which method a benchmark recommends, that the effect is carried by the backward reach a scored row requires, and that the resulting loss of supervision is predictable from two counts a custodian already holds. A benchmark reporting accuracy without availability reports a conditional quantity as though it were unconditional.

Several limits are worth stating rather than leaving to a referee. The injection experiment runs at one horizon with one injected gap length distribution, and its three donors are

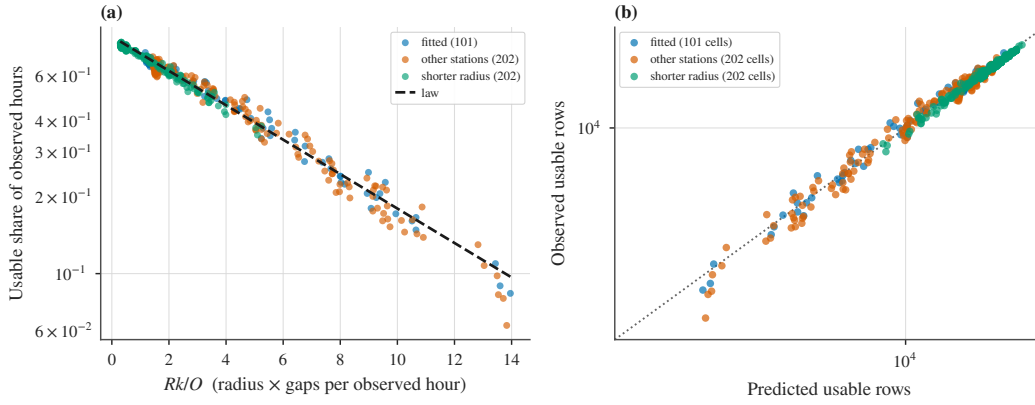


Fig. 3. The amplification law of (2). (a) Usable share of observed hours against Rk/O , fitted on 101 cells of one station's grid at $R = 192$. (b) Predicted against observed usable rows for all 505 cells. The held out sets are two distinct extrapolations, one across stations at the fitted radius and one across radii on the fitting station.

stations of a single archive covering one four year window and one weather regime. That makes it a station robustness check, able to falsify a station specific effect but not evidence of generality across cities. The comparative claim that the sequence tier suffers most does not survive an intersection union test and is not made, and the mediation result is established only on the record where the penalty was measured. The target series comes from a single monitor, so results describe that site rather than Dhaka as a whole. Ridge is structurally disadvantaged by a logarithmic target with raw scale history and should be read as a linear floor rather than a tuned competitor. Finally, inverting that logarithmic scale returns something nearer a conditional median, so every model here underpredicts the largest episodes. The bias is common to all of them, so between model comparisons remain valid.

VI. CONCLUSION

Record fragmentation is not a preprocessing detail. Holding a record fixed and manipulating only the arrangement of its absences costs the sequence tier 0.0394 skill at constant data volume, and that is the only family whose loss replicates across three stations. The mechanism is the backward reach a scored row demands. Quartering that reach removes 97 per cent of the penalty, and the supervision a record can support follows a closed form in observed hours, gap count and reach that predicts 404 held out cells at $R^2 = 0.982$. Price the reach before choosing it, because here the configured 168 hour reach bought no accuracy and cost fourteen points of the hours a forecast could be issued at all.

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